Does Noise-Adaptive Momentum Help? A Controlled Negative-Result Study of Adam-NV

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***Abstract*—Adaptive optimizers such as Adam use a fixed first-moment decay rate β_1. Adam-NV replaces it with a per-element β_1 that falls when a local gradient-noise score rises, so that noisy gradients are trusted less. We test this idea against Adam, AdamW, AdaBelief and the β_1-adaptive GONO optimizer on CIFAR-10 and CIFAR-100 with a ResNet-18, using paired seeds and decision rules fixed before the controlled runs. On clean data Adam-NV shows no detectable advantage over Adam (10 seeds: $+0.06$ pp, 95% CI $[-0.05, +0.16]$), under three non-stationary gradient-noise schedules, or on CIFAR-100. At learning rate 3×10^{-3}, an exploratory four-seed gain of $+0.20$ pp shrinks to $+0.17$ pp over ten seeds (95% CI $[+0.05, +0.29]$) and is not significant after multiplicity correction (Holm $p = 0.057$). Adam with a constant β_1 of 0.83 or 0.80 gains only $+0.06$ pp, and the data neither confirm nor exclude a small advantage of Adam-NV over these controls ($+0.12$ pp, CI $[-0.02, +0.25]$). Under 20% label noise, Adam-NV memorizes 3.9 pp more corrupted training labels than Adam in all eight seeds, no constant β_1 between 0.70 and 0.90 reproduces this, and letting β_1 rise instead (smooth response) lowers it 1.5 pp below Adam, although a constant β_1 of 0.95 does about as well (-1.0 pp, CI $[-2.2, +0.2]$). Permuting β_1 across the elements of each tensor keeps about 60–70% of the extra memorization, so most of the effect does not depend on the link between an element's β_1 and its own score, but a remainder of 1.1–1.5 pp does. Held-out accuracy is unchanged. Adam-NV costs about 17% more time per training step and twice the optimizer state, and it outperforms GONO by 0.76 pp. We conclude that noise-adaptive momentum gives no practically meaningful accuracy benefit in these settings (any gain at 3×10^{-3} is at most about 0.3 pp), while it measurably changes how strongly corrupted labels are fitted.**

***Index Terms*—adaptive momentum, Adam, gradient noise, label noise, memorization, negative results, optimizer evaluation**

I. Introduction

Momentum-based adaptive optimizers dominate deep-network training, and almost all of them fix the momentum coefficient at a default such as $\beta_1 = 0.9$ [1]. The coefficient controls how long past gradients influence the update, and the best value plausibly depends on how reliable recent gradients are. This motivates *adaptive momentum*: lower β_1 when gradients look unreliable, raise it when they look consistent.

Adam-NV is one such design. It estimates a per-element noise score from the first and second moment estimates already maintained by Adam and lowers β_1 elementwise when that score is high. The idea is cheap to state and attractive to deploy, but whether it helps is an empirical question that is easy to answer wrongly. A learned β_1 that settles below the default can improve results simply because a lower fixed β_1 would improve them, and small gains of a few tenths of a point are comparable to the seed-to-seed spread of typical benchmarks [2]–[4].

We therefore ask three questions. (i) Does Adam-NV improve accuracy over Adam and related optimizers under clean training, learning-rate changes, non-stationary gradient noise and a second dataset? (ii) If a gain appears, does it survive a control in which Adam uses a constant β_1 matched to the one Adam-NV actually learned? (iii) Does the adaptation change training behavior in ways accuracy does not reveal, for example in how strongly the model memorizes corrupted labels?

Contributions.

1) A paired-seed evaluation of Adam-NV against Adam, AdamW, AdaBelief and the β_1-adaptive GONO optimizer, with statistics reported as effect sizes with confidence intervals and Holm-adjusted p-values.
2) A matched constant-β_1 control and a ten-seed confirmation at learning rate 3×10^{-3}, where an exploratory four-seed gain shrinks to $+0.17$ pp (Holm $p = 0.057$) and is only partly reproduced by constant-β_1 Adam ($+0.06$ pp); the sample does not settle whether the remainder is real.
3) A label-noise result that the control cannot explain: Adam-NV memorizes 3.9 pp more corrupted labels than Adam, whereas constant β_1 from 0.90 down to 0.70 changes memorization by at most 0.8 pp; letting β_1 rise instead (smooth response) lowers it 1.5 pp below Adam, which a constant β_1 of 0.95 largely matches, and a shuffle control shows that most of the extra memorization survives breaking the element-wise link while a remainder of about 1.1–1.5 pp does not.
4) A measurement of the computational cost and a release of logs, scripts and decision rules.

II. Related Work

Adaptive optimizers. Adam [1] combines momentum with per-element step sizes from a running second moment. Later

variants repair its convergence behavior (AMSGrad [5]), stabilize early steps (RAdam [6]), add Nesterov momentum (Nadam [7]), decouple weight decay (AdamW [8]), or scale steps by the deviation of the gradient from its running mean (AdaBelief [9]). All keep β_1 fixed. Classical momentum [10], [11] is likewise used with a constant coefficient.

Adapting momentum. GONO [12] sets β_1 per parameter tensor from the cosine similarity of consecutive gradients, so that directionally consistent gradients receive more momentum. It is the closest published β_1-adaptive method to Adam-NV and is therefore our main adaptive baseline. The gradient noise scale [13] relates noise to the useful batch size, which motivates using a noise estimate to steer optimizer hyperparameters.

Label noise and memorization. Deep networks fit random labels [14], learn clean patterns before corrupted ones [15], and early stopping helps under label noise [16]. We use label corruption as a controlled setting in which the *noise-adaptive* part of an optimizer has a chance to matter, and we measure memorization directly.

Evaluating optimizers. Optimizer comparisons depend strongly on tuning and on seed variance [2]–[4], [17]. We follow this line by pairing runs by seed, reporting uncertainty, and adding controls for the quantity that the proposed method adapts.

III. Methods

A. Adam-NV

All operations are elementwise. Let g_t be the gradient at step t, m_t and v_t the first and second moment estimates, and $\beta_{1,t}$ the momentum coefficient. Adam-NV computes a moment-based noise proxy from the moments of the previous step,

$$s_t = \frac{\max\big(v_{t-1} - m_{t-1}^2,\, 0\big)}{v_{t-1} + \lambda\,|m_{t-1}| + \epsilon}, \tag{1}$$

smooths it with a warm-up weight and an exponential average,

$$\rho_t = \alpha\,\rho_{t-1} + (1-\alpha)\,\frac{t}{t+k}\,s_t, \tag{2}$$

and maps it to a momentum coefficient that can only fall below the base value β_{base}:

$$\beta_{1,t} = \beta_{\min} + (\beta_{\text{base}} - \beta_{\min})\max(1 - \gamma\rho_t,\, 0)\,. \tag{3}$$

The moment updates use the time-varying coefficient,

$$m_t = \beta_{1,t}\,m_{t-1} + (1-\beta_{1,t})\,g_t, \tag{4}$$

$$v_t = \beta_2\,v_{t-1} + (1-\beta_2)\,g_t^2. \tag{5}$$

Because $\beta_{1,t}$ varies, the usual factor $1-\beta_1^t$ is replaced by $1-P_t$, where $P_t = \max\big(\prod_{s\le t}\beta_{1,s},\, 10^{-7}\big)$, and the update is

$$\theta_{t+1} = \theta_t - \eta\,\frac{m_t/(1-P_t)}{\sqrt{v_t/(1-\beta_2^t)} + \epsilon}. \tag{6}$$

State is initialized to zero ($m_0 = v_0 = \rho_0 = 0$, $P_0 = 1$). The implementation supports parameter groups and L2 weight decay (default 0), skips parameters without gradients, and does not support sparse gradients, AMSGrad or fused kernels. Unless stated otherwise we use $\beta_{\text{base}} = 0.9$, $\beta_{\min} = 0.1$, $\beta_2 = 0.999$, $\gamma = 1$, $\alpha = 0.95$, $k = 100$, $\lambda = 0.1$ and $\epsilon = 10^{-8}$, without tuning. The numerator estimates a second central moment with exponential averages and the ratio is bounded by one. The proxy is not a calibrated variance: because v has a much longer memory than m, it also rises when gradient magnitudes shrink during training, and we did not validate it against a direct measurement of gradient noise. With $\gamma = 0$ and β_{base} fixed, the method equals Adam apart from the clamp on P_t, which is reached after about 150 steps for $\beta_1 = 0.9$ and changes the bias-correction factor by at most 10^{-7}. A variant (*smooth* response) lets $\beta_{1,t}$ also rise above β_{base} toward 0.999 when the score is low; the variant above is the *distrust* response. Two further variants lower the second-moment decay β_2 or give the noise score its own decay (*decoupled*); they are used only in the non-stationary noise experiments and are compared with Adam-NV itself.

B. Baselines and controls

Adam uses $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$ and no weight decay. **AdamW** uses weight decay 10^{-4}. **AdaBelief** uses the standard form without rectification. **GONO** sets β_1 per tensor from the cosine similarity c_t of consecutive gradients, $\beta_1 = \text{clip}(\beta_{\text{base}}(1+\lambda_G c_t), \beta_{\text{lo}}, \beta_{\text{hi}})$ with $\lambda_G = 0.4$, $\beta_{\text{lo}} = 0.5$ and $\beta_{\text{hi}} = 0.99$. We re-implemented GONO following the paper, with the per-tensor cosine and cosine $\epsilon = 10^{-10}$ of the authors' released code. Our ϵ placement in the Adam step follows the paper and PyTorch, whereas the released code places it differently; on synthetic gradients the two implementations differ by at most 2.5×10^{-6} in parameter value, and they agree to 10^{-15} once both differences are matched.

The key control is **constant-β_1 Adam**, the Adam-NV code with $\gamma = 0$ and β_1 fixed at a chosen value. In every setting we first take the time average of Adam-NV's logged β_1 and set the control to that value. The logged statistic is a mean over parameter tensors that gives every tensor the same weight; the element-weighted mean is higher (Section V-E), so the matched constants are lower than the controller's element-level average, and the range of constants we test (0.70 to 0.95) brackets both. Agreement with torch Adam for a fixed β_1 is checked by a unit test.

IV. Experimental Setup

Data and model. We use CIFAR-10 and CIFAR-100 [18] (50,000 training and 10,000 test images) with a ResNet-18 [19] whose stem is adapted to 32×32 inputs. Training uses random crop with padding 4 and horizontal flip, cross-entropy loss, batch size 128, 60 epochs (≈23,400 steps) and cosine-annealed learning rate [20]. Learning rate is 10^{-3} unless stated. The CIFAR test set serves as the validation set and every result is the value at fixed epochs, with no checkpoint selection. Software was PyTorch [21] 2.10.0 (CUDA 12.8) with Python 3.12.13 on Tesla T4 GPUs, with deterministic mode enabled.

Experiments. Table I lists every experiment. Gradient noise, when injected, is norm-scaled Gaussian noise [22]: for every parameter tensor, a random direction of unit Frobenius norm scaled by $\sigma\|g\|_F$ is added to its gradient, so with $\sigma = 2.0$ the perturbation is twice as large as the gradient. It is applied during the middle third of training (epochs 21–40, *phased*), ramped linearly from zero to σ over training (*ramp-up*), or in alternating five-epoch noisy and clean blocks (*intermittent*). For label noise, 20% of the training labels are replaced once by a random different class; the corrupted subset is determined by the run seed, so every optimizer sees the same corruption within a seed, and the test labels are untouched. At the end of every epoch we pass the training set through the network in evaluation mode and record the fraction of corrupted examples predicted as their *wrong* label (the memorization rate) and the accuracy on uncorrupted examples; we average the last five epochs to reduce epoch-to-epoch fluctuation. This measures how strongly corrupted labels are fitted; it is not a measure of generalization.

Metrics and statistics. The primary accuracy metric is the mean validation accuracy over the last five epochs; we also report epochs to 90% accuracy and the validation-loss area under the curve (sum over 60 epochs). Runs are paired by seed. For each contrast we report the mean paired difference in percentage points (pp), a t-based 95% confidence interval, the paired t-test p-value, the exact two-sided sign-flip p-value (with $n = 4$ it cannot fall below 0.125), and Holm-adjusted p-values [23] within each experiment's family of contrasts. All differences are computed from unrounded per-seed values, so they can differ by 0.01 pp from differences of the rounded means shown in the tables. Inference rests on the paired t interval with Holm adjustment; the sign-flip test is a distribution-free check. We use three labels: a *detected* difference (Holm-adjusted $p < 0.05$), *no detectable* difference (the interval includes zero), and *practical equivalence*, which we claim only if the whole 95% interval lies within ± 0.15 pp. No comparison in this paper currently meets the equivalence bar, so "no detectable difference" never means "equal".

Chronology and test-set use. The clean benchmark and the learning-rate experiment came first and were exploratory. The learning-rate gain motivated the constant-β_1 control, and a memorization difference seen in a label-noise diagnostic motivated the label-noise controls; these follow-up experiments used decision rules written in run books before their runs. No hyperparameter, stopping or checkpoint choice used the test set, but later experiments were chosen after earlier test-set results were seen, so p-values that pool exploratory seeds are optimistic. The ten-seed extension at learning rate 3×10^{-3} adds six new seeds (4–9) whose results are also reported separately from the exploratory seeds 0–3. Logs record library versions, arguments and SHA-256 hashes of the source files, and a repeat of two label-noise runs with the final code reproduced the earlier logs exactly.

V. Results

A. Clean benchmark

Table II shows that Adam-NV, Adam and AdamW are statistically indistinguishable on clean CIFAR-10. Adam-NV exceeds Adam by 0.06 pp (95% CI $[-0.05, +0.16]$, Holm $p = 0.29$), a smaller gap than the $+0.12$ pp seen with the first four seeds, which is what seed noise would predict. AdamW's gap is similar ($+0.08$ pp). GONO and AdaBelief are both clearly lower than Adam (-0.70 pp, CI $[-0.82, -0.59]$, and -0.71 pp, CI $[-0.80, -0.62]$) and than Adam-NV ($+0.76$ pp, CI $[+0.62, +0.90]$, and $+0.77$ pp, CI $[+0.63, +0.90]$; 10 of 10 seeds each; all Holm $p < 10^{-5}$). Holm adjustment covers the six contrasts of this experiment (Adam-NV, AdamW, GONO and AdaBelief against Adam, and Adam-NV against GONO and AdaBelief). A GONO variant that uses Adam-NV's exact running-product bias correction instead of the paper's base correction gives $92.38 \pm 0.12\%$, within 0.06 pp of the base version ($p = 0.27$) and still 0.76 pp below Adam, so GONO's deficit is not caused by its bias-correction choice.

B. Learning-rate sensitivity and the constant-β_1 control

Table IV shows the only accuracy gain we found. At learning rate 3×10^{-3} Adam-NV exceeds Adam by $+0.20$ pp (95% CI $[+0.11, +0.28]$, higher in 4 of 4 seeds, Holm $p = 0.014$ within the four learning-rate contrasts); at 3×10^{-4} the difference is -0.02 pp. GONO is 1.54 pp below Adam at 3×10^{-3} and reaches 90% accuracy about 16 epochs later, and ties Adam at 3×10^{-4}. Adam-NV's logged β_1 averaged 0.83 at 3×10^{-3} and never exceeded its base value 0.9; the lowest value was 0.57. These logged statistics are means over the elements of each parameter tensor, then over tensors, so the minimum is the lowest tensor mean and not the lowest element.

Constant-β_1 control and seed extension. The four-seed controls in Table IV gained $+0.17$ and $+0.20$ pp over Adam, as much as Adam-NV. Those four seeds, however, generated the hypothesis, and Adam happened to score low on them (92.94% against 93.07% on seeds 4–9). We therefore ran seeds 4–9 for all four arms under a rule fixed beforehand (Table III). Over ten seeds Adam-NV exceeds Adam by $+0.17$ pp (95% CI $[+0.05, +0.29]$, higher in 9 of 10 seeds, raw $p = 0.011$, Holm $p = 0.057$ over the five contrasts of this experiment). Its gain is similar on the exploratory seeds ($+0.20$ pp) and on the new seeds alone ($+0.16$ pp, CI $[-0.08, +0.39]$, 5 of 6 seeds). The constant controls do not hold up: they gain $+0.06$ and $+0.07$ pp over Adam on ten seeds and -0.02 and -0.03 pp on the new seeds alone. Adam-NV minus the controls is $+0.116$ pp (CI $[-0.021, +0.253]$) and $+0.105$ pp (CI $[-0.035, +0.246]$), with Adam-NV higher in 8 of 10 seeds in both cases (Holm $p = 0.35$ and 0.37). By the pre-specified rule the outcome is inconclusive: the controls reproduce less than half of the gain, but the interval of the Adam-NV-minus-control difference includes zero. The data are compatible with an Adam-NV advantage of up to about 0.25 pp over a constant β_1, and with none; with the observed spread, about 24 paired

TABLE I
OVERVIEW OF EXPERIMENTS. ALL USE RESNET-18, 60 EPOCHS AND BATCH SIZE 128 (TIMING EXCEPTED).

Experiment	Setting	Arms	Seeds	Decision rule fixed beforehand
Clean benchmark	CIFAR-10, lr 10^{-3}	Adam, Adam-NV, AdamW, GONO (and a GONO variant), AdaBelief	10	no (descriptive)
Learning-rate sensitivity	CIFAR-10, lr 3×10^{-3} and 3×10^{-4}	Adam, Adam-NV, GONO	4	no (exploratory)
Constant-β_1 control	CIFAR-10, lr 3×10^{-3}	Adam, Adam-NV, Adam with $\beta_1 = 0.83$ and 0.80	10 (4 exploratory, 6 new)	yes
Non-stationary gradient noise	phased, ramp-up, intermittent; $\sigma = 2.0$	Adam, Adam-NV and β_2 variants	4 / 4 / 8	no (Holm-adjusted contrasts)
Controller direction, CIFAR-100	distrust vs smooth response; CIFAR-100 clean and intermittent	Adam-NV modes; Adam vs Adam-NV	4; 8 / 4	no (Holm-adjusted contrasts)
Label noise, control	20% label noise, lr 10^{-3}	Adam, Adam-NV, constant $\beta_1 = 0.81$	8	yes
Label noise, dose-response	same	constant $\beta_1 = 0.78$ and 0.70	8	yes
Label noise, response direction	same	Adam-NV with smooth response	8	yes
Label noise, β_1 shuffle	same	Adam-NV with β_1 permuted (fresh each step, or one fixed permutation); no-shuffle canary	8 (canary 2)	yes
Optimizer overhead	Tesla T4, random batch of 128	Adam, Adam-NV, AdamW, AdaBelief	10 repeats	not applicable

TABLE II
CLEAN CIFAR-10, LEARNING RATE 10^{-3} (MEAN $\pm$ SD OVER SEEDS).

Optimizer	Seeds	Acc. (%)	Epochs to 90%	Val.-loss AUC
Adam	10	93.15 ± 0.11	19.0	26.31
Adam-NV	10	93.20 ± 0.15	18.6	26.39
AdamW	10	93.22 ± 0.15	18.4	26.43
GONO	10	92.44 ± 0.14	25.4	30.39
AdaBelief	10	92.44 ± 0.17	24.3	26.67

One GONO log lost its final row; the run was repeated and reproduced the other 59 epochs exactly.

TABLE III
TEN-SEED CONFIRMATION AT LEARNING RATE 3×10^{-3} (LAST-FIVE ACCURACY; SEEDS 0–3 EXPLORATORY, SEEDS 4–9 NEW). HOLM OVER THE FIVE CONTRASTS.

Arm	Seeds	Acc. (%)	Δ (pp)	95% CI	Holm p	Seeds above
Adam	10	93.02 ± 0.16	—	—	—	—
Adam-NV	10	93.19 ± 0.16	$+0.17$	$[+0.05, +0.29]$	0.057	9/10
const. $\beta_1 = 0.83$	10	93.07 ± 0.21	$+0.06$	$[-0.11, +0.22]$	0.82	7/10
const. $\beta_1 = 0.80$	10	93.08 ± 0.18	$+0.07$	$[-0.11, +0.24]$	0.82	8/10
Adam-NV − const. 0.83	10	—	$+0.12$	$[-0.02, +0.25]$	0.35	8/10
Adam-NV − const. 0.80	10	—	$+0.11$	$[-0.04, +0.25]$	0.37	8/10

TABLE IV
LEARNING-RATE SENSITIVITY AND CONSTANT-β_1 CONTROLS (FOUR SEEDS, LAST-FIVE ACCURACY, MEAN $\pm$ SD).

LR	Optimizer	Acc. (%)	Δ vs Adam (pp)	95% CI	Holm p
3×10^{-3}	Adam	92.94 ± 0.11	—	—	—
	Adam-NV	93.13 ± 0.14	$+0.20$	$[+0.11, +0.28]$	0.014
	GONO	91.40 ± 0.25	-1.54	$[-2.03, -1.05]$	0.009
	const. $\beta_1 = 0.83$	93.11 ± 0.15	$+0.17$	$[-0.03, +0.36]$	0.28
	const. $\beta_1 = 0.80$	93.14 ± 0.16	$+0.20$	$[-0.11, +0.52]$	0.39
3×10^{-4}	Adam	92.33 ± 0.08	—	—	—
	Adam-NV	92.30 ± 0.16	-0.02	$[-0.31, +0.26]$	1.00
	GONO	92.18 ± 0.31	-0.15	$[-0.77, +0.47]$	1.00

Holm families: the learning-rate experiment (Adam-NV and GONO against Adam at each learning rate, four contrasts) and the control experiment (each control against Adam, and Adam-NV against each control, four contrasts) are adjusted separately.

TABLE V
ADAM-NV MINUS ADAM UNDER INJECTED GRADIENT NOISE (LAST-FIVE ACCURACY).

Schedule	Seeds	Adam-NV	Adam	Δ (pp) [95% CI]	Holm p
Phased	4	92.86	92.75	$+0.11$ $[-0.35, +0.56]$	1.00
Ramp-up	4	92.97	92.95	$+0.02$ $[-0.22, +0.26]$	1.00
Intermittent	8	92.37	92.53	-0.16 $[-0.41, +0.09]$	0.51

seeds would be needed to resolve a difference of 0.1 pp. The final-epoch analysis gives the same picture (Adam-NV minus Adam $+0.16$ pp, Holm $p = 0.13$).

C. Non-stationary gradient noise

Table V shows that none of the three schedules produces a significant difference between Adam-NV and Adam, and the largest gap favors Adam in the intermittent schedule (-0.16 pp, CI $[-0.41, +0.09]$). Two follow-up families vary Adam-NV's second-moment decay and are compared with Adam-NV itself, not with Adam. In the first, a lower decay $\beta_2 \in \{0.99, 0.95\}$ is compared with the default 0.999 for each of four schedules (eight contrasts, four seeds each). One contrast is Holm-significant: under ramped noise, $\beta_2 = 0.95$ is 0.16 pp *below* the default (CI $[-0.22, -0.10]$, Holm $p = 0.028$; the exact sign-flip p is 0.125, its minimum for four seeds). The largest gain is $+0.44$ pp for $\beta_2 = 0.95$ under intermittent noise (CI $[+0.06, +0.82]$, raw $p = 0.035$, Holm $p = 0.25$), and the remaining six contrasts have Holm $p \geq 0.64$. In the second family, a separate decay for the noise score is compared with the coupled value (eight contrasts); the smallest Holm p is 0.068 (ramped noise, decay 0.99, $+0.13$ pp).

D. Controller direction and CIFAR-100

Letting β_1 also rise above 0.9 (smooth response) does not help: the default distrust response is nominally ahead by

TABLE VI
CONTROLLER DIRECTION (DISTRUST MINUS SMOOTH) AND ADAM-NV MINUS ADAM ON CIFAR-100.

Comparison	Seeds	Δ (pp)	95% CI	Holm p
Direction, clean	4	$+0.10$	$[-0.23, +0.42]$	0.42
Direction, const. noise	4	$+0.09$	$[-0.07, +0.25]$	0.36
Direction, intermittent	4	$+0.29$	$[+0.01, +0.57]$	0.14
CIFAR-100, clean	8	-0.29	$[-0.62, +0.04]$	0.15
CIFAR-100, intermittent	4	-0.01	$[-0.64, +0.63]$	0.97

$+0.10$ pp (clean), $+0.09$ pp (constant noise) and $+0.29$ pp (intermittent noise; CI $[+0.01, +0.57]$), but no Holm-adjusted value is below 0.14 (Table VI). On CIFAR-100 Adam-NV is 0.29 pp below Adam on clean data (eight seeds, CI $[-0.62, +0.04]$, Holm $p = 0.15$, Adam higher in six of eight seeds) and tied under intermittent noise (-0.01 pp, four seeds).

E. Label noise and memorization

Both Adam and Adam-NV reach about 87.5% accuracy at epoch $\approx$22 and then decline to roughly 78.8% as the corrupted labels are fitted. Held-out accuracy at the end is the same across all arms (78.5–78.9%), and so is the peak. Memorization, however, differs (Table VIII, Fig. 1). Adam-NV fits $83.5 \pm 0.5\%$ of the corrupted labels against $79.6 \pm 1.0\%$ for Adam, a difference of $+3.91$ pp (95% CI $[+3.11, +4.71]$), with Adam-NV higher in all eight seeds (exact sign-flip $p = 0.0078$, the minimum for eight seeds; Holm $p < 0.001$).

The constant-β_1 control matters here. Adam with a constant β_1 equal to the time average of Adam-NV's logged per-tensor mean (0.81) gains only $+0.64$ pp (CI $[-0.40, +1.67]$). Because Adam-NV's logged β_1 drifts from 0.87 at the first epoch to about 0.78 late in training, with the lowest per-tensor mean around 0.6 (Fig. 2; element-level values were logged only in the shuffle-control runs, Table VII), we also ran constant β_1 of 0.78 and 0.70. They gain $+0.77$ and $+0.83$ pp, neither significant, and Adam-NV stays 3.14 pp (CI $[+2.16, +4.13]$, Holm $p = 0.0005$) and 3.08 pp (CI $[+1.92, +4.24]$, Holm $p = 0.0012$) above them, in 8 of 8 seeds each. The dose-response of memorization to constant β_1 is flat within noise over $0.9 \to 0.7$. Under the pre-specified rule the outcome is that the effect is not explained by any constant β_1 tested.

Direction of the adaptation. The smooth response lets β_1 rise above 0.9 toward 0.999 when the score is low (logged mean β_1 0.92, range 0.90–0.99). It gives the opposite result: it fits $78.11 \pm 0.89\%$ of the corrupted labels, 1.47 pp *fewer* than Adam (CI $[-2.40, -0.55]$, Holm $p = 0.007$, lower in 8 of 8 seeds), and 5.38 pp fewer than the distrust response (CI $[+4.55, +6.21]$, 8 of 8 seeds). Validation accuracy is again unchanged within noise (smooth minus Adam -0.42 pp, CI $[-0.95, +0.12]$). Under the rule fixed for that experiment the outcome was that the distrust direction drives the effect, but that rule did not include constants above 0.9, which we tested next. The experiment also does not show whether the element-wise link between β_1 and each element's own score is needed.

Constant β_1 above 0.9. The smooth response also raises the average β_1 (logged mean 0.92), so we ran constants of 0.92 and 0.95 under a rule fixed beforehand. A constant β_1 of 0.95 fits $78.56 \pm 1.06\%$ of the corrupted labels, 1.03 pp fewer than Adam (CI $[-2.21, +0.16]$, lower in 6 of 8 seeds), which is statistically indistinguishable from the smooth response (smooth minus constant 0.95: -0.45 pp, CI $[-1.32, +0.43]$). A constant of 0.92 shows no reduction ($+0.44$ pp, CI $[-0.82, +1.71]$) and a larger spread across seeds (SD 1.79 against 0.9 to 1.3 for the other arms); the smooth response is 1.91 pp below it (CI $[-3.52, -0.30]$, Holm $p = 0.10$). The pre-specified rule gives "adaptation adds" for 0.92 and "reproduces" for 0.95, so the overall outcome is partial. Pooling all constant-β_1 runs from 0.70 to 0.95, memorization falls as β_1 rises (slope -5.7 pp per unit of β_1, $r = -0.38$, $p = 0.008$; descriptive only). We therefore read the smooth-response reduction as consistent with a higher average β_1 and not as a property of the adaptation, whereas the extra memorization of the distrust response is not mirrored by any constant we tested.

Which part of the controller matters. To separate the element-wise link from the distribution of β_1, we permuted Adam-NV's per-element β_1 across the elements of each parameter tensor before applying it, either with a fresh permutation every step or with one fixed permutation per tensor; the distribution of values is unchanged, their link to each element's own score is destroyed. A run through the same wrapper without permutation reproduced the two existing Adam-NV logs bit for bit (maximum difference 0 over all 15 logged columns). Both permuted variants keep most of the effect: they fit 2.40 pp (CI $[+1.20, +3.60]$) and 2.81 pp (CI $[+2.18, +3.44]$) more corrupted labels than Adam, 61% and 72% of Adam-NV's $+3.91$ pp, in 8 of 8 seeds each. They remain below Adam-NV by 1.51 pp (CI $[+0.93, +2.08]$, Holm $p = 0.002$) and 1.10 pp (CI $[+0.39, +1.81]$, Holm $p = 0.016$), and the two permutations do not differ detectably from each other. By the pre-specified rule the outcome is partial: most of the effect does not need the element-wise link, and a remainder of about 1.1 to 1.5 pp does. Validation accuracy is unchanged within noise (-0.20 pp for both variants, intervals including zero).

The runs also give the element-level distribution of the controller's β_1 (Table VII). It is much closer to the base value than the per-tensor mean suggests: the element-weighted mean stays between 0.88 and 0.89, the median between 0.887 and 0.898, and at epoch 60 only 5% of the elements lie below 0.845 and 1% below 0.818; fewer than 0.07% fall below 0.7. The per-tensor mean of 0.78 in Fig. 2 is lower because small tensors, such as normalization parameters, reach much lower values and count as much as the large convolution tensors in that average. The constant controls therefore lowered β_1 for every element more than Adam-NV does for most elements, which makes a matched constant a conservative comparison; because we tested constants from 0.70 to 0.95, which bracket the element-level mean, this does not change the conclusion that no constant we tested reproduces the effect.

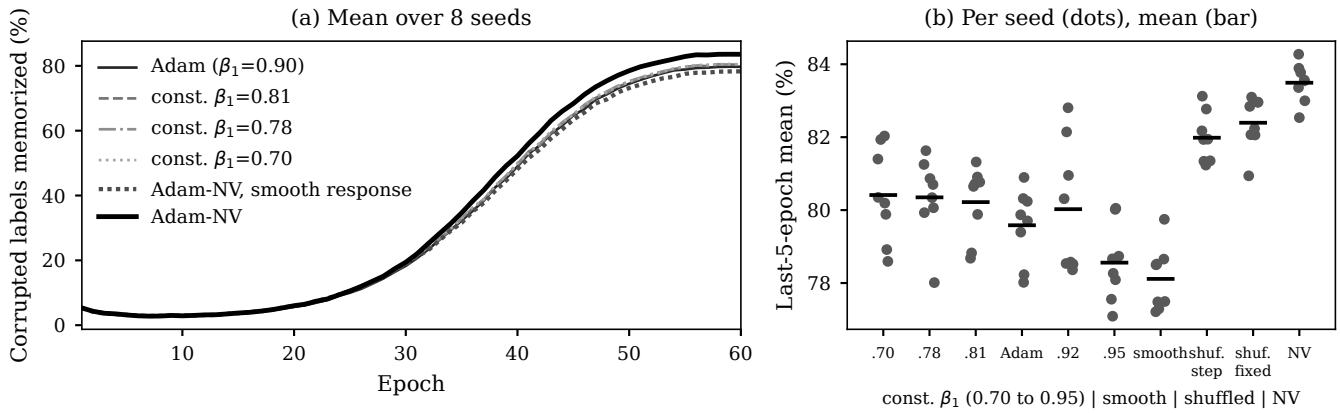


Fig. 1. Memorization of corrupted labels under 20% label noise. (a) Mean over eight seeds per epoch. (b) Last-five-epoch value per seed (dots) and mean (bar); Adam-NV is separated from every constant-β_1 control, the smooth response lies below Adam, and the two shuffled variants lie between Adam and Adam-NV.

TABLE VII
ELEMENT-LEVEL DISTRIBUTION OF ADAM-NV'S β_1 UNDER LABEL NOISE (MEAN OF TWO RUNS, ALL 11.2 MILLION ELEMENTS).

Epoch	Mean	Median	5% quantile	1% quantile	Minimum	Below 0.7 (%)
1	0.891	0.894	0.876	0.857	0.601	0.01
20	0.893	0.898	0.871	0.846	0.486	0.03
40	0.885	0.891	0.850	0.824	0.406	0.02
60	0.881	0.887	0.845	0.818	0.436	0.02

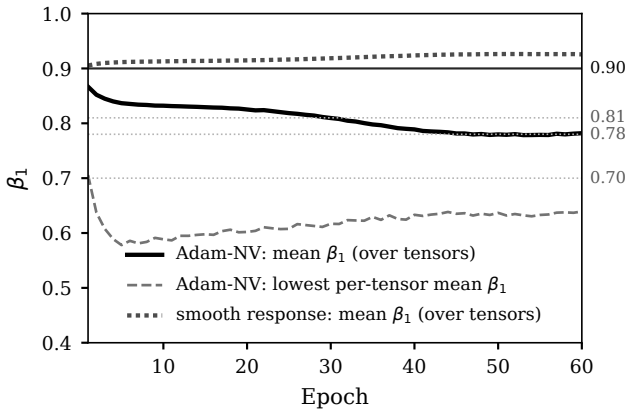


Fig. 2. Adam-NV's momentum coefficient under label noise: mean over parameter tensors (each tensor's mean over its elements) and seeds, and the mean lowest per-tensor value, and, for the smooth response, the mean over tensors, against the constant-β_1 controls. Element-level values are in Table VII.

F. Computational cost

We timed Adam, Adam-NV, AdamW and AdaBelief on a Tesla T4 with a random input batch of 128 images in 20 rounds. In every round each optimizer is timed once, in a freshly randomized order, with 50 warm-up and 200 measured steps and GPU synchronization around each timed region; we report medians and ratios paired within a round, with bootstrap 95% intervals over rounds. The optimizer step takes 27.3 ms for Adam-NV against 3.8 ms for Adam (7.2 times as long, interval $[7.14, 7.18]$). The end-to-end training step (zero-grad, forward, backward, update) takes 126.5 ms against 108.0 ms, an increase of 17.1% (interval $[16.9, 17.5]$%); AdamW adds 0.4% and AdaBelief 2.8%. The end-to-end difference (18.5 ms) is smaller than the step-only difference (23.5 ms) because the forward and backward pass is 5.1 ms faster in the Adam-NV runs (99.0 against 104.2 ms). This persists under randomized order (Adam's total time varies by less than 1 ms across positions in a round), and we have not explained it. Adam-NV keeps four per-element state buffers (first moment, second moment, smoothed score and running product) against two for Adam, so its optimizer state is twice as large, 170.5 MiB against 85.3 MiB for our 11,173,962-parameter ResNet-18, and its peak allocated GPU memory is 85.6 MiB higher (914.1 against 828.5 MiB, $+10.3$%). The intervals describe round-to-round variation within one session on one GPU, and the figures apply to our unfused PyTorch implementation.

VI. DISCUSSION

What the evidence supports. Across clean training, three learning rates, three noise schedules and two datasets, Adam-NV never exceeds Adam by an amount that survives correction for multiple comparisons (the smallest Holm p is 0.057, at learning rate 3×10^{-3}). The only candidate, $+0.17$ pp at 3×10^{-3}, is not significant after multiplicity correction on ten seeds and is only partly reproduced by constant-β_1 Adam ($+0.06$ pp), so we cannot tell whether the remainder reflects noise-awareness or chance; its upper confidence limit is about 0.3 pp. Because the constant controls recovered only about a third of the point estimate, we do not claim that a lower fixed β_1 is a substitute, and we did not run a budget-matched tuning study. Against the other β_1-adaptive method, Adam-NV is better at 10^{-3} and 3×10^{-3} and tied at 3×10^{-4}, so adapting momentum through a noise score is a better choice

TABLE VIII

LABEL NOISE (20%), LEARNING RATE 10^{-3}, EIGHT SEEDS. MEMORIZATION IS THE LAST-FIVE-EPOCH MEAN FRACTION OF CORRUPTED TRAINING EXAMPLES PREDICTED AS THEIR WRONG LABEL; CONTRASTS ARE PAIRED BY SEED.

Arm	Memorization (%)	Val. acc. (%)	Clean-subset acc. (%)	Δ vs Adam (pp)	95% CI	Seeds above Adam
Adam, $\beta_1 = 0.90$	79.58 ± 1.01	78.79	98.73	—	—	—
Constant $\beta_1 = 0.81$	80.22 ± 0.99	78.80	98.74	$+0.64$	$[-0.40, +1.67]$	7/8
Constant $\beta_1 = 0.78$	80.35 ± 1.11	78.85	98.75	$+0.77$	$[-0.45, +1.99]$	6/8
Constant $\beta_1 = 0.70$	80.41 ± 1.30	78.53	98.77	$+0.83$	$[-0.57, +2.22]$	6/8
Constant $\beta_1 = 0.92$	80.03 ± 1.79	78.58	98.74	$+0.44$	$[-0.82, +1.71]$	5/8
Constant $\beta_1 = 0.95$	78.56 ± 1.06	78.61	98.70	-1.03	$[-2.21, +0.16]$	2/8
Adam-NV	$\mathbf{83.49 \pm 0.54}$	78.84	98.84	$\mathbf{+3.91}$	$[+3.11, +4.71]$	8/8
Adam-NV, smooth response	78.11 ± 0.89	78.37	98.72	-1.47	$[-2.40, -0.55]$	0/8
Adam-NV, β_1 shuffled each step	81.98 ± 0.69	78.59	98.80	$+2.40$	$[+1.20, +3.60]$	8/8
Adam-NV, β_1 shuffled once (fixed)	82.40 ± 0.73	78.59	98.78	$+2.81$	$[+2.18, +3.44]$	8/8

Adam-NV minus constant β_1: $+3.27$ pp (0.81), $+3.14$ pp (0.78), $+3.08$ pp (0.70); all CIs exclude 0 (Holm $p \leq 0.0012$; Holm families are the three contrasts of the control experiment and the four of the dose-response experiment). Adam-NV minus smooth response: $+5.38$ pp ($[+4.55, +6.21]$, Holm $p < 0.001$ within the two direction contrasts). Adam-NV minus shuffled-each-step: $+1.51$ pp ($[+0.93, +2.08]$, Holm $p = 0.002$); minus shuffled-once: $+1.10$ pp ($[+0.39, +1.81]$, Holm $p = 0.016$); shuffled-once minus shuffled-each-step: $+0.41$ pp ($[-0.53, +1.35]$); Holm over the five shuffle contrasts.

than adapting it through gradient direction in our setting, although neither beats a fixed β_1.

What the label-noise result does and does not show. The extra memorization is the one effect that a constant β_1 cannot reproduce, and the smooth-response ablation shows that the direction of the adaptation matters: raising β_1 where the score is low lowers memorization below Adam's level. A constant β_1 of 0.95 reproduces that reduction within noise, so the reduction is not clearly specific to adaptation, whereas the increase under the distrust response is. The shuffle control narrows down the cause. Permuting β_1 across the elements of each tensor keeps about 60 to 70% of the extra memorization, so most of the effect comes from the distribution of β_1 values the controller produces (a small lowering for many elements and a large lowering for a few) and not from each element's β_1 matching its own score; a remainder of about 1.1 to 1.5 pp does depend on that match. We did not test why the distribution itself raises memorization. The effect did not change held-out accuracy at the end of training, so we do not claim it is harmful for generalization here, and an early-stopped model would not show a difference.

Threats to validity. (i) All results use one architecture family and CIFAR-scale data; optimizer rankings can change with scale. (ii) Hyperparameters were not tuned for any optimizer, including Adam-NV; a tuned Adam-NV could differ. (iii) Several experiments use four seeds, so confidence intervals are wide (half-widths of 0.2 to 0.3 pp, against about 0.1 pp for ten seeds) and the sign-flip test cannot go below 0.125; "no detectable difference" is not equivalence. (iv) The CIFAR test set doubles as validation, although no model selection uses it. (v) Label noise is symmetric and uniform at one rate and learning rate. (vi) Holm families were defined by us per experiment and are stated in the table notes. (vii) We re-implemented GONO; its ϵ placement differs from the authors' released code in a way we measured on synthetic gradients but did not evaluate on CIFAR. (viii) Timing was measured on one GPU type. (ix) The logged β_1 statistics are per-tensor means; the element-level distribution is available only from two label-noise runs, and the matched constants were set from the per-tensor statistic. (x) The smooth response also raises the average β_1, and a constant β_1 of 0.95 reproduces its reduction within noise, so that reduction is not clearly specific to adaptation; the 0.92 arm also had an unusually large spread across seeds (SD 1.79). (xi) AdaBelief and GONO were run with literature-default settings and no tuning (AdaBelief with $\epsilon = 10^{-8}$, no weight decay, no rectification); AdaBelief is known to be sensitive to ϵ, so its lower accuracy here is not a general ranking.

VII. CONCLUSION

We evaluated noise-adaptive momentum with matched controls and found no detectable accuracy improvement of Adam-NV over Adam in the tested settings. Its one candidate gain (at most about 0.3 pp at 3×10^{-3}, not significant after correction) is only partly reproduced by a lower constant β_1, it costs about 17% more per step, and under label noise it memorizes more corrupted labels than Adam or any constant-β_1 control, without changing held-out accuracy, and letting β_1 rise lowers it below Adam's, which a constant β_1 of 0.95 largely matches. For work on adaptive hyperparameters, these experiments suggest reporting the adapted quantity's trajectory and comparing against a constant control matched to it, and, under label noise, measuring memorization directly. Code, logs, run books and decision rules are available at https://github.com/abiroh-p/Adam-NV. [TODO: clean the repository layout and reconcile the stored source files with the hashes recorded in the logs before this link is published]

ACKNOWLEDGMENT

The authors thank their supervisor, Ms. Repshika Pradhan, for guidance. [TODO: confirm acknowledgment wording and any funding or compute statement]

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